

KESHAV GOEL

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EDUCATION

International Institute of Information Technology, Hyderabad (IIIT-H)

B.Tech in Computer Science and Engineering | **CGPA: 9.35/10.0**

July 2024 – May 2028

Class XII (CBSE): **96.2%** (2024) | Class X (CBSE): **97.0%** (2022)

ACHIEVEMENTS & EXTRACURRICULARS

Mathematical Aptitude: Secured **Rank 597 (Top 0.03%)** in JEE Mains among 1.4+ million candidates; **Rank 2634** in JEE Advanced, demonstrating elite mathematical and logical problem-solving capabilities.

Computational Intelligence: Solved **550+ algorithmic problems** on Codeforces (max rating 1332), LeetCode, and CSES, excelling in advanced data structures, combinatorial logic, and execution optimization.

Leadership & Collaboration: Manage logistics and cross-functional operations for **DebSoc** and the **Cultural Taskforce**, demonstrating strong communication and stakeholder management under tight execution deadlines.

Academic Excellence: Awarded **Dean's List (Semesters 1 and 2)** at IIIT Hyderabad, ranking in the **top 5%** of the undergraduate cohort.

EXPERIENCE

Sprinklr

Software Engineering Intern | Distributed Systems, Enterprise Software

Gurgaon, India

Upcoming: Summer 2027

- Secured a highly competitive internship offer to engineer enterprise-scale software solutions.

Nanocare Health

Software Engineering Intern | Python, FastAPI, Backend Architecture

Hyderabad, India

January 2026 – April 2026

- Collaborated effectively within a cross-functional team to architect a highly robust backend API, processing and serving **407,000+ structured data points** for secure institutional data retrieval.
- Implemented extensive unit testing and integration of analytical de-identification algorithms (Presidio) to guarantee 100% GDPR compliance in a fast-paced environment.

PROJECTS

Trading Strategy Simulation & Backtesting | Python, Object-Oriented Design

[GitHub](#)

- Engineered proprietary **trading strategy simulation software** modeling an ATM straddle on NIFTY indices, specifically optimized for high-volume financial data ingestion and algorithmic trade execution.
- Built robust pipelines for **large-scale data acquisition and visualization**, processing tick data second-by-second to generate reproducible real-time trade management logs and comprehensive risk metrics.

Scalable Distributed Systems & Reliable UDP | C, TCP/IP, Multithreading

[DFS](#) | [UDP](#)

- Architected a fault-tolerant distributed storage system utilizing **multi-threaded applications**, fine-grained **sentence-level mutex locking**, and FNV-1a hashing for **O(1) lookups** to manage concurrent accessibility.
- Engineered custom **network programming** middleware by developing a Reliable UDP protocol over TCP/IP concepts, implementing precise packet recovery, sequencing, and retransmission logic to guarantee data integrity.

High-Performance Low-Latency Compute | C++, CUDA, Triton, AVX2

[BLAS](#) | [Profiling](#)

- Designed and tested extremely fast **C++ BLAS kernels** utilizing AVX2 SIMD parallel instructions, executing rigorous mathematical optimizations to achieve up to **5.0x computational speedups (6.59 GFLOPS)**.
- Employed sophisticated command-line profiling tools (gprof, perf) to isolate and systematically resolve a severe **59% cache-miss bottleneck**, ensuring robust execution for latency-sensitive execution flows.

OS Kernel Architecture & POSIX Shell | C, Linux/Unix, IPC

[Kernel](#) | [C-Shell](#)

- Demonstrated strong **Linux/Unix working knowledge** by extending core xv6 kernel data structures, developing custom memory management (demand paging) and thread schedulers to optimize hardware utilization.
- Programmed a POSIX-compliant UNIX shell featuring n-length pipelines, background job control, and **signal-safe real-time atomic operations** for flawless inter-process communication.

TECHNICAL SKILLS & COURSEWORK

Languages: C++, C, Python, SQL, JavaScript

Core Capabilities: Data Structures & Algorithms, Object-Oriented Design, Network Programming (TCP/IP), Multithreading, Low-Latency Architecture, Computational Intelligence, Distributed Systems

Performance & OS: Linux/Unix, SIMD/AVX2, CUDA, Triton, CPU Profiling (perf, gprof), Git, POSIX API

Coursework: Probability & Statistics, Design & Analysis of Algorithms, Software Programming for Performance, Computer Systems Organisation, Design & Analysis of Software Systems, Operating Systems & Networking